



UKCP09: Probabilistic projections of wind speed

UKCP09 additional product



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Probabilistic projections of wind speed: Executive summary

- Probabilistic projections of changes in 30-year mean wind speeds relative to 1961–1990 have been produced for UKCP09, based on the same methodology previously used to produce other variables available online. As for other UKCP09 variables, the probabilistic climate projections of changes in wind speed reflect current scientific understanding.
- This report presents examples of the changes for the 2050s and 2080s. The range of projected changes in summer wind speed for the 2050s covers both positive and negative changes but is generally slightly skewed towards negative changes. This is indicated by small reductions at most UK locations in the 50% probability level, between 0 ms^{-1} and -0.2 ms^{-1} . A windspeed of 0.2 ms^{-1} is about 0.4 knots which is small compared with the typical magnitude of summer mean wind speed of about 7–10 knots over much of England and Northern Ireland, and 7–14 knots over much of Scotland and Wales.
- In winter for the 2050s, the ranges of projected changes in wind speed are approximately symmetric about near-zero change. The largest uncertainty ranges are over Scotland, where the 10–90% probability range is typically -0.5 ms^{-1} to 0.5 ms^{-1} . This range equates to -1 to 1 knots, however these changes in 30-year mean winter wind speeds are small compared with the typical winter averages for present day climate, which are 10–14 knots over the Lowlands and 18–24 knots over the Highlands.
- As for other variables in UKCP09, the projected changes are derived principally from ensembles of perturbed variants of the Met Office Hadley Centre (HadCM3) model. These show a variety of changes, due to a spread of forced responses to increased greenhouse gas concentrations, augmented considerably by natural climate variability. The forced changes are influenced by several different factors, and vary between different ensembles, and between individual ensemble members. While the changes in large scale atmospheric circulation patterns over the North Atlantic and Europe show some coherent features traceable to basic aspects of the response to increased greenhouse gases, there is little evidence of a consistent response in surface wind speed over the UK. The spread due to uncertainties in the forced response is typically larger in winter than in summer, but is always smaller than the spread due to natural variability. These combined factors provide much of the total uncertainty in the projections. Where the changes show a small shift in the balance of probabilities towards lower or higher wind speeds, this

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arises from the net impact of a variety of physical mechanisms captured in the HadCM3 ensembles, and in a small ensemble of 12 international climate models used to complement the sampling of model uncertainties achieved via the HadCM3 simulations. However, these shifts are always small in the context of the observed climatological mean wind speed.

- The results (as for all UKCP09 products) are subject to the caveat that the current generation of climate models used to produce them could be missing a key process liable to change the projections. In the case of wind speed, some recent research suggests that including a better resolved stratosphere in the climate model could alter the projected changes over Europe, particularly in winter. However, the evidence should be regarded as preliminary until a wider range of relevant modelling studies are available. The next set of climate model projections being generated around the world for the IPCC Fifth Assessment Report will include several models containing improved representations of the effects of the stratosphere on surface climate. Following provision and assessment of these, it will be possible to assess the implications for the current results in a more concrete fashion.

1 Background

The probabilistic projection methodology in UKCP09 involves sampling climate modelling uncertainties by combining results from perturbed variants of the HadCM3 configuration of the Met Office global climate model with projections from an ensemble of alternative international climate models. However, as discussed in the UKCP09 climate projections science report (Murphy *et al.* 2009), it was not possible to provide probabilistic projections of future changes for certain variables (soil moisture, latent heat flux, snowfall rate and wind speed). In the case of wind speed, the required data was not available from the other global climate models. This is because an accurate calculation of monthly mean wind speed in a climate model has to be based on the accumulated values of wind speed at individual model timesteps, and such diagnostics were not available for the other global climate models. As an alternative, Brown *et al.* (2009) provided advice on changes in wind speed based on the range of responses from an ensemble of 11 variants of our own regional climate model that were used in the production of UKCP09. Brown *et al.* (2009) concluded that the regional model simulations did not sample the full range of possible outcomes, but that they could be used to obtain a set of plausible future realisations of surface winds (mean values and time series characteristics), once corrected for biases in the simulated historical values.

However, after further investigation since the launch of UKCP09, we have found that the required monthly mean wind speeds for the ensemble of other climate models can be well approximated by accumulating values of wind speed calculated from the daily means of the westerly and northerly components of wind. This was demonstrated by estimating projected monthly wind speed changes from time series of daily wind components for an ensemble of variants of HadCM3, and comparing against the correct monthly wind speed changes based on timestep-accumulated values. Figure 1 shows that the biases from this approach are below 0.1 ms^{-1} , which is small compared to the uncertainties in the projected changes in section 3. Daily wind components are available for the 7 of the 12 alternative climate models, allowing monthly wind speeds to be estimated as described above. Monthly wind speed values for the remaining 5 models were estimated from other variables, using inter-variable correlations obtained from the ensemble of alternative models, following Murphy *et al.* (2009).

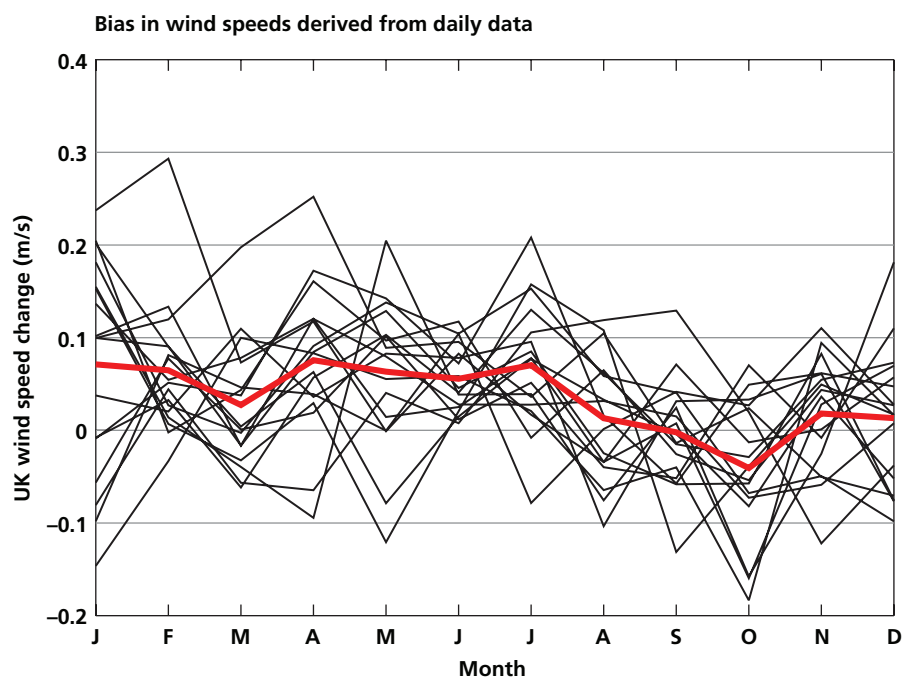


Figure 1: Ensemble mean bias (thick line) in monthly mean wind speed calculated from daily means of wind components for 17 configurations of HadCM3 (thin lines).

2 Large scale changes in circulation

Future changes in mean wind speed, and indeed in other climatological metrics relating to storms or anticyclones (see Annex 6 of Murphy *et al.* 2009), can potentially be influenced by several aspects of the forced response to increases in greenhouse gases or aerosols. In particular, a number of factors likely to affect future changes in mid-latitude cyclones have been identified in previous research (e.g. McDonald, 2010). For example, increases in atmospheric moisture content in a warmer atmosphere provide an increased source of energy for the intensification of storms through condensation and precipitation, once the storms have formed. In the upper troposphere, the warming is larger in the tropics than in polar regions, tending to provide more energy for storm development. In winter, on the other hand, the zonally averaged surface warming at high latitudes exceeds that found at lower latitudes, tending to reduce the available energy. Also, the land warms more than the oceans. In northern hemisphere winter, this reduces the temperature contrasts which provide energy for the genesis of storms off the eastern coasts of the continents. Regarding the UK and western Europe, another factor is the relative minimum in the projected warming typically found in central regions of the North Atlantic Ocean, particularly in model simulations including a dynamic ocean component, since these generally predict a reduction in the northward transport of warm water from the tropics associated with the Atlantic Meridional Overturning Circulation (AMOC – see Annex 5 of Murphy *et al.* 2009). In some (though not all) global model projections, this leads to a southward shift in the storm track downstream of the consequent changes in the north–south sea surface temperature gradient. The projected changes in wind and storm characteristics over the UK in any particular model simulation will therefore depend on a balance between several competing factors, and will in addition be influenced by the unpredictable effects of natural climate variability.

For these reasons, the changes in circulation simulated in our ensembles of perturbed variants of HadCM3 show considerable variability over the UK, which are reflected in a wide spread of projected changes in mean surface wind speed. In our largest ensemble, which simulates the equilibrium response to doubled carbon dioxide concentrations using a simplified ocean, the ensemble-averaged circulation changes in the troposphere (e.g. Figure 2a) show modest increases in the mean westerly flow during winter in the Atlantic/European sector, between about 40 and 60 degrees north. These increases are consistent with regional increases in the north–south temperature gradient in the upper troposphere in this latitude band, associated with the enhanced upper level warming simulated in tropical regions (see above). However, the ensemble-averaged increases in surface wind speed over the UK are small (not exceeding 0.2 ms^{-1}). Furthermore, there is a spread in the ensemble of projected changes: The forced component

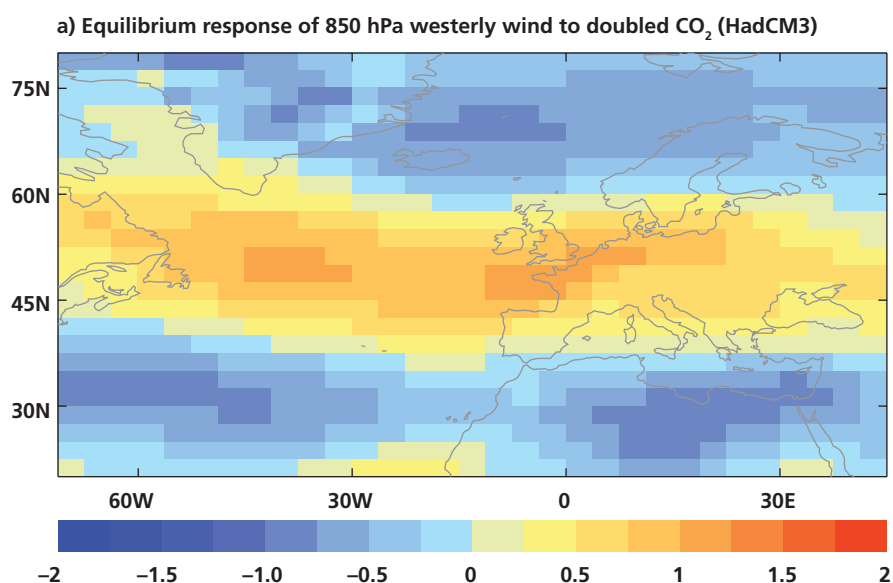
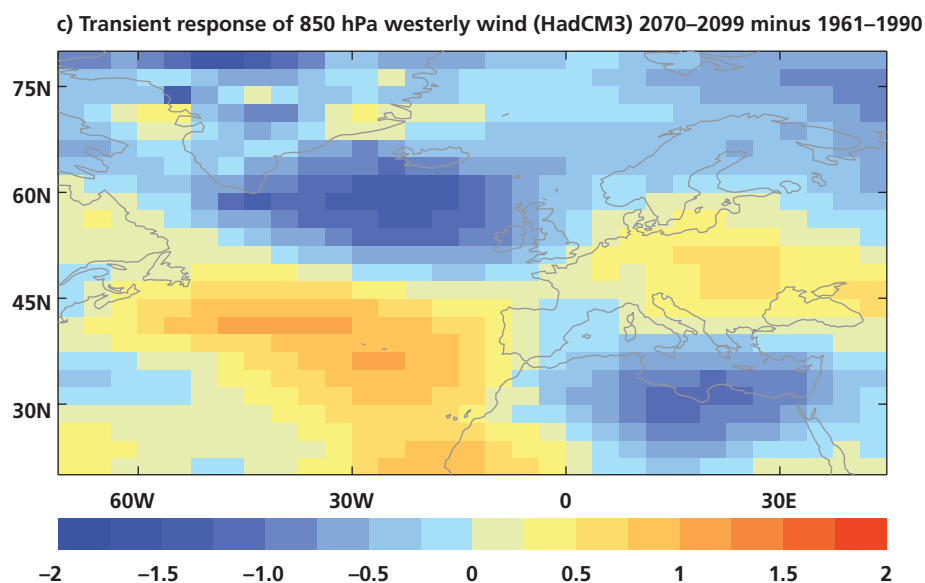
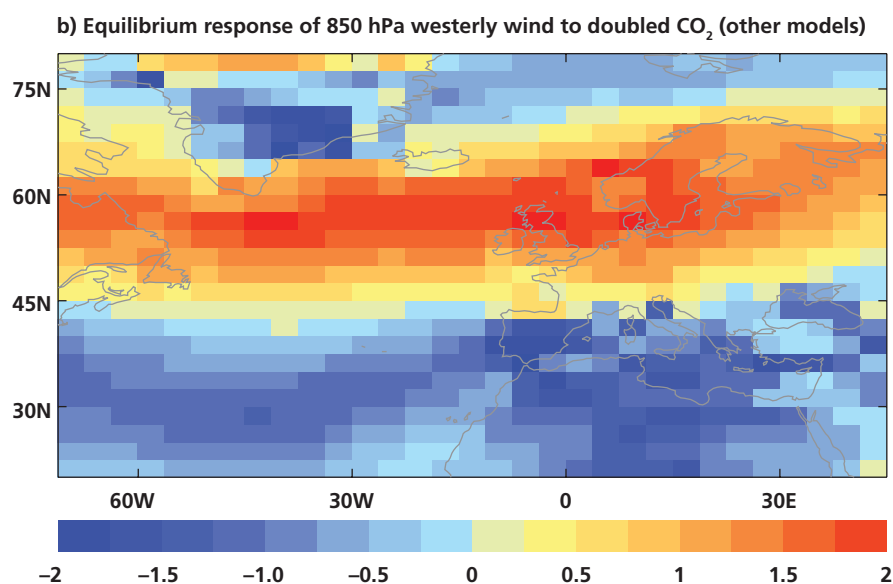


Figure 2: Comparison of changes in winter 850 hPa westerly wind speed for a) ensemble mean of equilibrium response to doubled CO₂ of perturbed variants of HadCM3, b) ensemble mean of equilibrium response to doubled CO₂ from other models, and c) ensemble mean of transient response (2070–2099 minus 1961–1990) of variants of HadCM3.



of the change is negative in some ensemble members, and uncertainties are augmented considerably by the influence of natural climate variability, which is larger than the spread of forced changes. The corresponding ensemble of 12 alternative climate models shows a similar large-scale pattern of changes in atmospheric winds in its ensemble average (Figure 2b), however the band of increased winds is displaced slightly to the north compared with the HadCM3 ensemble. This is probably because the enhancement in surface warming at high latitudes, which tends to reduce the energy available for storm development, is somewhat more pronounced in the HadCM3 ensemble. Over the UK, the multi-model ensemble therefore shows somewhat larger increases in lower tropospheric winds over Scotland and northern England, resulting in slightly larger ensemble-averaged increases in the mean surface wind speed. Again, however, there is a considerable spread in the responses of individual ensemble members.

In order to produce probabilistic projections for specific 21st century periods, further simulations of time-dependent climate change are required, in addition to the simulations of equilibrium climate change discussed above. In particular, the time-dependent simulations included a dynamical ocean module, and therefore account for the effects of ocean transport changes. In these simulations, also consisting of an ensemble of variants of HadCM3 (see Murphy *et al.* 2009), a smaller surface warming is predicted in the North Atlantic ocean relative to the response over Greenland and the eastern seaboard of the north American continent. This reduced warming is consistent with reductions in northward heat transport via the AMOC, and reduces the energy for the genesis and development of storms. This leads to an ensemble average response showing reductions in the mean westerly wind over the Atlantic and western Europe, relative to the equilibrium simulations (see Figure 2c). Therefore, over the UK, this mechanism tends to offset the modest ensemble-averaged increases in surface wind speed found in the simulations of equilibrium climate change. The probabilistic projections reflect the combined sampling of these competing mechanisms of change and their associated uncertainties, achieved by pooling the equilibrium and time-dependent simulations. The results for winter therefore tend to show distributions of changes centred on a near-zero response, as described below.

In summer, ensemble average patterns of change in the simulations of equilibrium climate change are consistent with a poleward shift in the summer storm track found in previous climate model projections (Yin, 2005). This is associated with the impacts of an enhanced equator-to-pole temperature gradient in the upper troposphere. The patterns of changes in atmospheric westerly winds are broadly similar between the HadCM3 and multi-model ensembles (Figure 3a cf 3b), showing a band of increases extending eastwards from north-eastern Canada across the Atlantic, and over most of the UK and northern Europe, with a band of reduced westerlies to the south. Over the UK, this results in small increases in the ensemble averaged changes in surface wind speed to the north, with smaller increases (in the HadCM3 ensemble) or small decreases (in the multi-model ensemble) over southern England, which lies on the northern edge of the band of reduced westerlies. In the ensemble of time-dependent HadCM3 projections a similar mean pattern of changes is found, however the northward shift is slightly more pronounced than in the equilibrium simulations. Again this is consistent with the effects of reductions in the AMOC, which limits the surface warming in the north Atlantic, and hence tends to increase further the equator-to-pole contrast in the atmospheric response. The impact of ocean heat transport changes therefore tends to offset the modest increases in westerly winds found over most of the UK in the equilibrium simulations, since the zonal band of increased westerlies is displaced to the north (Figure 3c). As in winter, the forced

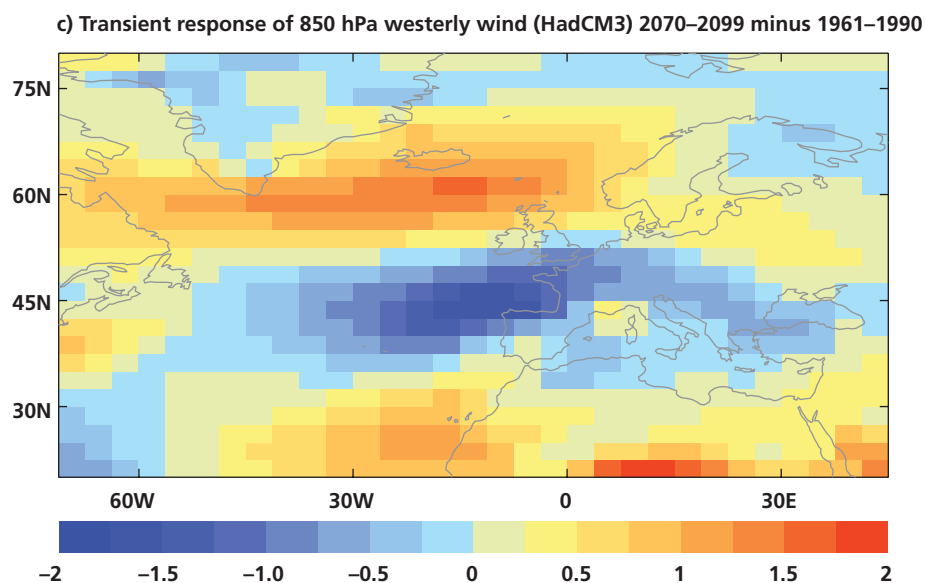
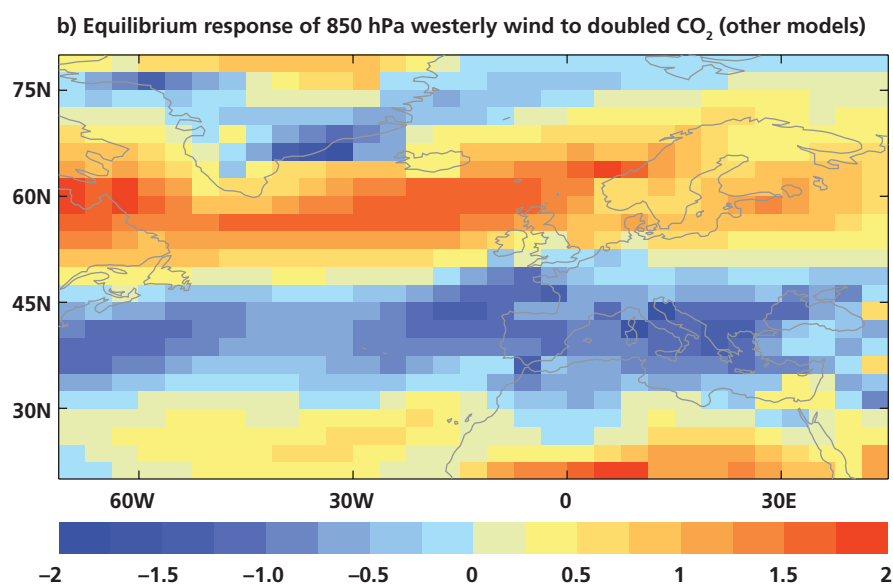
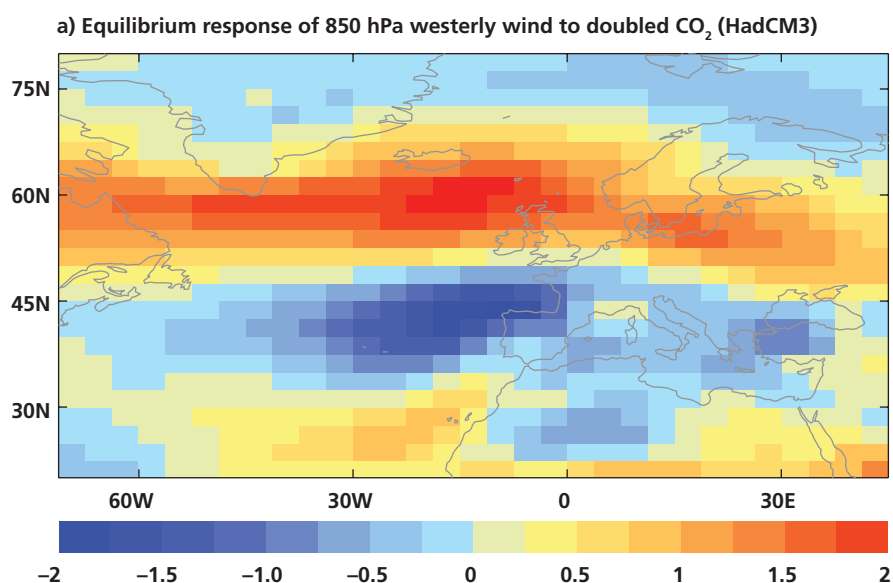


Figure 3: Comparison of changes in summer 850 hPa westerly wind speed for a) ensemble mean of equilibrium response to doubled CO₂ of perturbed variants of HadCM3, b) ensemble mean of equilibrium response to doubled CO₂ from other models, and c) ensemble mean of transient response (2070–2099 minus 1961–1990) of variants of HadCM3.

responses in surface wind speed corresponding to these changes in atmospheric circulation are small. The forced responses also vary between ensemble members, and natural variability again adds a considerable additional component of uncertainty.

A basic assumption of our methodology for probabilistic projections is that our largest ensemble of HadCM3 variants, used to simulate equilibrium climate change as described above, provides a good first order estimate of the spread of responses consistent with modelling uncertainties. This ensemble is then augmented by use of the multi-model ensemble, in order to sample the effects of plausible variations in model structure not considered in the HadCM3 ensemble (see Chapter 3 of Murphy *et al.* 2009), and by further global and regional model ensembles simulating time-dependent climate change. If the contribution from the multi-model ensemble is too dominant (see Section 3.2.10 of Murphy *et al.* 2009, then its potential lack of robustness (being based on a small set of simulations assembled on an opportunity basis) could render the probabilistic projections unreliable. It is important to check this for wind speed, as for variables previously provided in UKCP09.

In general, the HadCM3 ensemble shows a wide spread of changes, with little evidence of a systematic signal when averaged across the ensemble members. Where the spread of responses does show a (small) systematic component, this is found to be consistent with the larger scale patterns of circulation change (as discussed above in the cases of summer and winter changes). In the multi-model ensemble, the regional differences in circulation changes relative to the HadCM3 (see Figure 2 and related discussion) do alter the projected wind speed changes over the UK to a limited degree. In general, the systematic component of the wind speed changes across members of the multi-model ensemble tends to be slightly larger than from the HadCM3 ensembles (though less so in summer), but remains modest compared with the spread of outcomes implied by the HadCM3 ensemble. Therefore, we conclude that the multi-model ensemble plays its expected role of modifying the information provided by the HadCM3 ensembles to account for the effects of structural model uncertainties not sampled in the latter. These impacts are modest, and are consistent with plausible uncertainties in how different physical mechanisms combine to produce changes in wind speed at the regional scale. We therefore conclude that production of probabilistic projections based on the combined information from the HadCM3 and multi-model ensembles is justified.

3 Probabilistic projections

Like the other variables in UKCP09, the probabilistic projections of wind speed were produced by combining results from the various model simulations outlined above with a set of observational constraints to produce the projections, using the statistical methodology described in Chapter 3 of Murphy *et al.* (2009). Projections were produced for the same set of emissions scenarios, regions, times of year and future periods described in section 1.2 of Murphy *et al.* (2009). Here we provide a few illustrative examples.

Figure 4 shows that the range of anomalies in summer wind speed for the 2050s covers both positive and negative changes but is slightly skewed towards negative changes, reflecting the combination of factors discussed earlier. The 50% probability level shows a tendency towards a small reduction in wind speed, with changes over most of the UK being between -0.2 ms^{-1} and 0 ms^{-1} . A windspeed of 0.2 ms^{-1} is about 0.4 knots which is small compared with the typical magnitude of summer mean wind speed of about 7–10 knots over much of England and Northern Ireland, and 7–14 knots over much of Scotland and Wales (Jenkins *et al.* 2008). In the 2050s there is very little change in the projections with emission scenario. By the 2080s (see Figure 5) the 10% probability level is more negative than that of the 2050s with values most negative along the west coast of England and Wales. For the 10% probability level, the minimum values are more negative for the higher emissions scenario.

For winter in the 2050s (see Figure 6), the ranges of anomalies in wind speed are quasi-symmetric about near-zero change. The 10–90% probability range is typically around -0.5 ms^{-1} to 0.5 ms^{-1} with a small variation in magnitude across the different emission scenarios. The spread in changes in wind speed is greatest over Scotland, with somewhat larger ranges of uncertainty of -0.7 ms^{-1} to 0.75 ms^{-1} are seen in the high emission scenario compared with the low emission scenario (about -0.4 ms^{-1} to 0.5 ms^{-1}). This range is roughly -0.8 to 1 knot and is relatively small compared with observed values of 10–14 knots over the lowlands of Scotland and 18–24 over much of the Highlands. We note that the spread of possible changes shown by the probability distributions is strongly influenced by natural climate variability. However, the spread is somewhat wider than that implied by natural variability alone (this is also true of the changes in other seasons), despite the lack of significant shifts in the values of the 50% probability level. This is partly due to the simulation of a spread of plausible forced responses in the ensembles of model projections, as discussed in Section 2. Uncertainties arising from the statistical techniques used to convert the model simulations into probabilistic projections also contribute to the spread of outcomes.

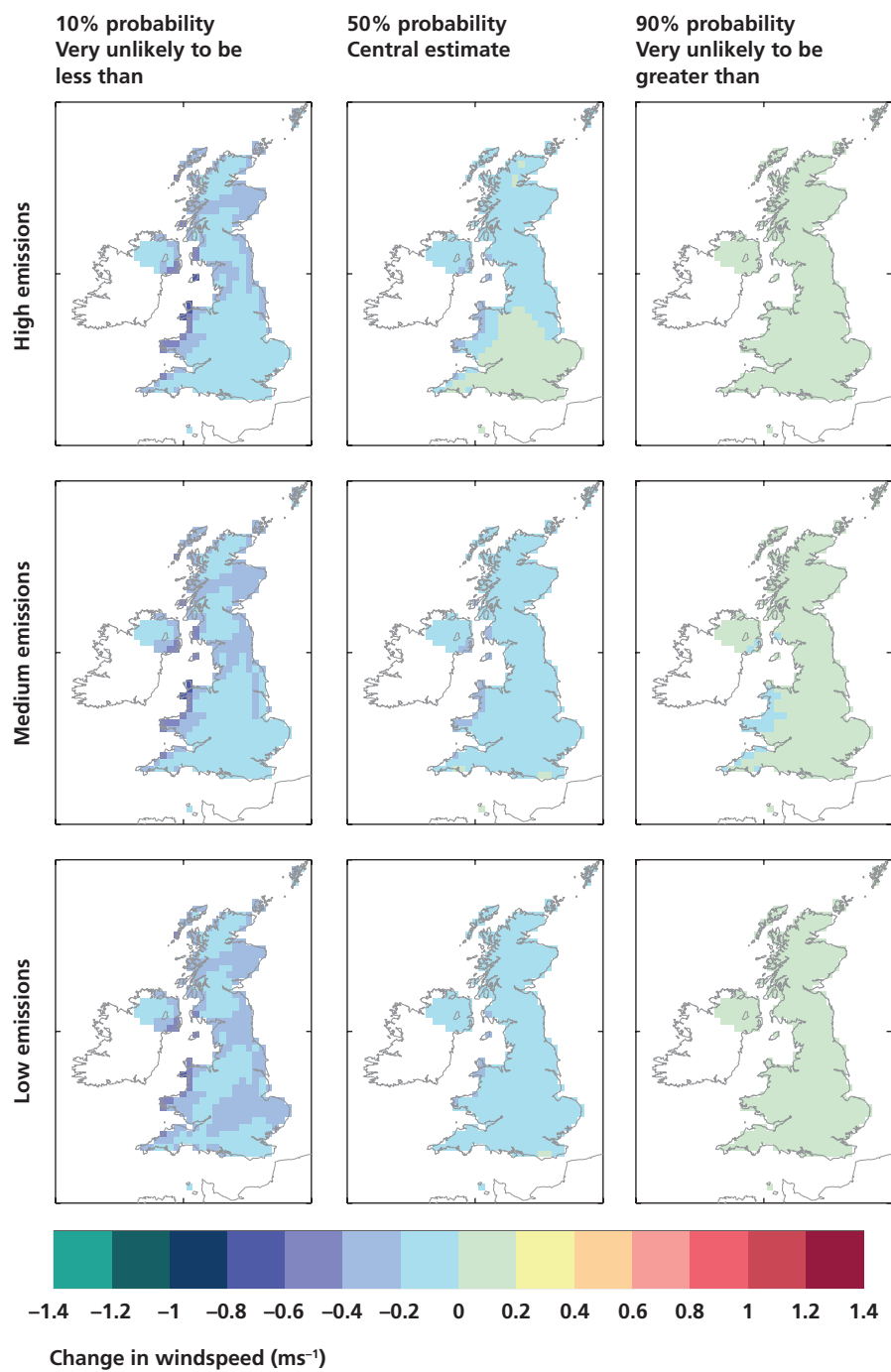


Figure 4: 10, 50, and 90% probability level maps of changes in summer wind speed (ms^{-1}) for the 2050s (2040–2069) relative to 1961–1990 for the Low (bottom row), Medium (middle row), and High (top row) emission scenarios.

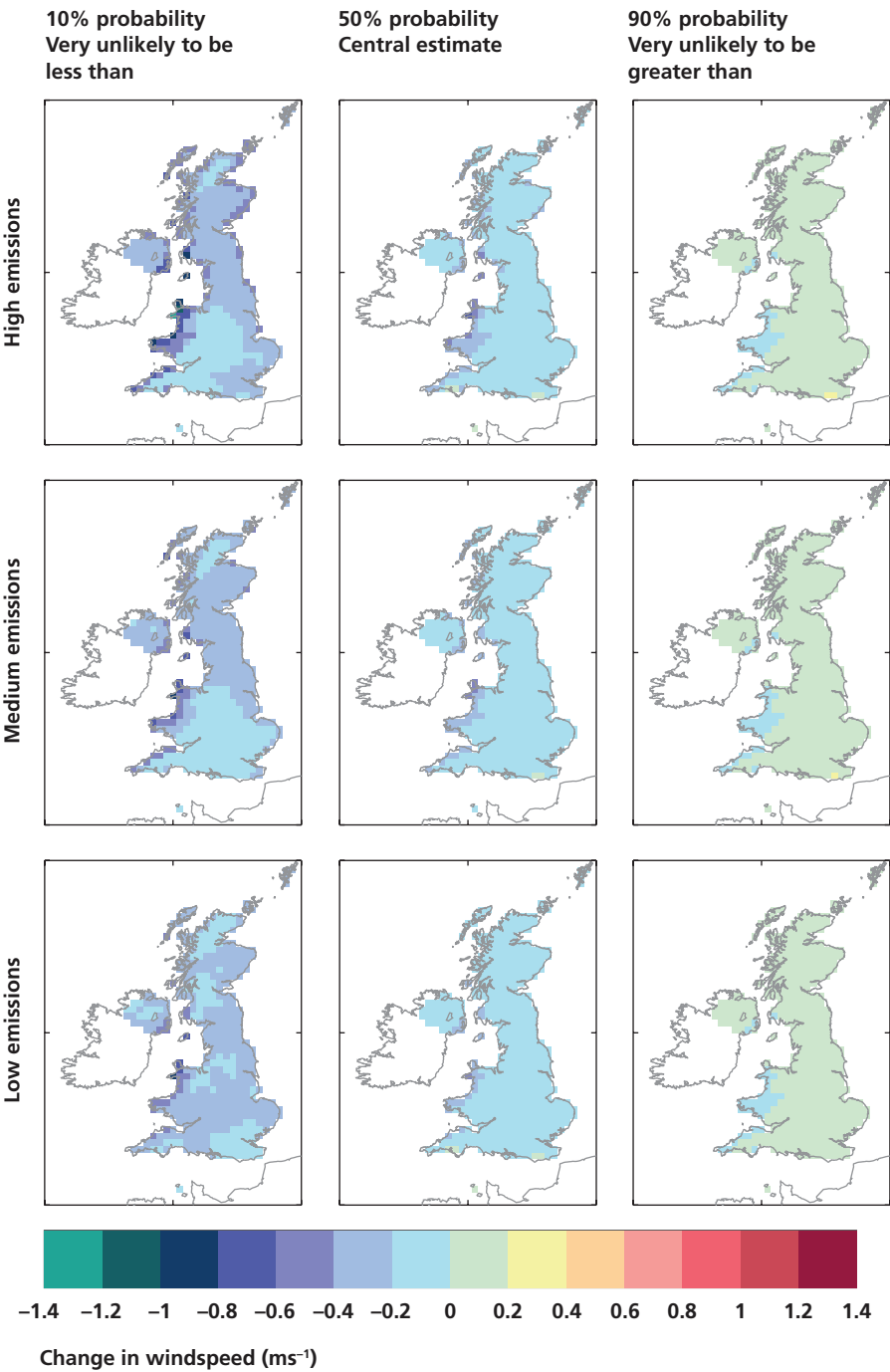


Figure 5: 10, 50, and 90% probability level maps of changes in summer wind speed (ms⁻¹) for the 2080s (2070–2099) relative to 1961–1990 for the Low (bottom row), Medium (middle row), and High (top row) emission scenarios.

In the 2080s (Figure 7) the 50% probability level shows small reductions in wind speed of about -0.1 ms^{-1} over the UK, except for the Highlands where small increases are found. The results again reflect the range of competing mechanisms and their associated uncertainties discussed earlier (see Jenkins *et al.* 2008).

We note that the changes in 30 year averages of wind speed presented here are only one of several possible metrics relating to future regional characteristics of the atmospheric circulation and its variability. Annex 6 of Murphy *et al.* (2009) reports changes in a number of diagnostics relating more specifically to variability associated with storms and anticyclones over the UK, though the general conclusion remains that it is difficult to identify consistent signals of increase or decrease.

All probabilistic climate projections in UKCP09 reflect the current understanding of the climate system as represented in the current state-of-the-art models, and it is possible that processes missing from all these climate models might affect the projections. For wind speed, this is particularly pertinent, as a recent study (Huebener *et al.*, 2007) suggests that the forced response of wind speeds over Europe could be significantly affected by how well the stratosphere is resolved in the climate models. Scaife *et al.* (2005) demonstrates that the simulated characteristics of low frequency variability associated with the North Atlantic Oscillation could also be improved. However, further research is needed to confirm the effects of troposphere–stratosphere interactions on future projections of European climate. The next set of projections being produced for the IPCC Fifth Assessment Report will include a number of climate models with a better resolved stratosphere. Once these become available, it will be possible to make a firmer assessment.

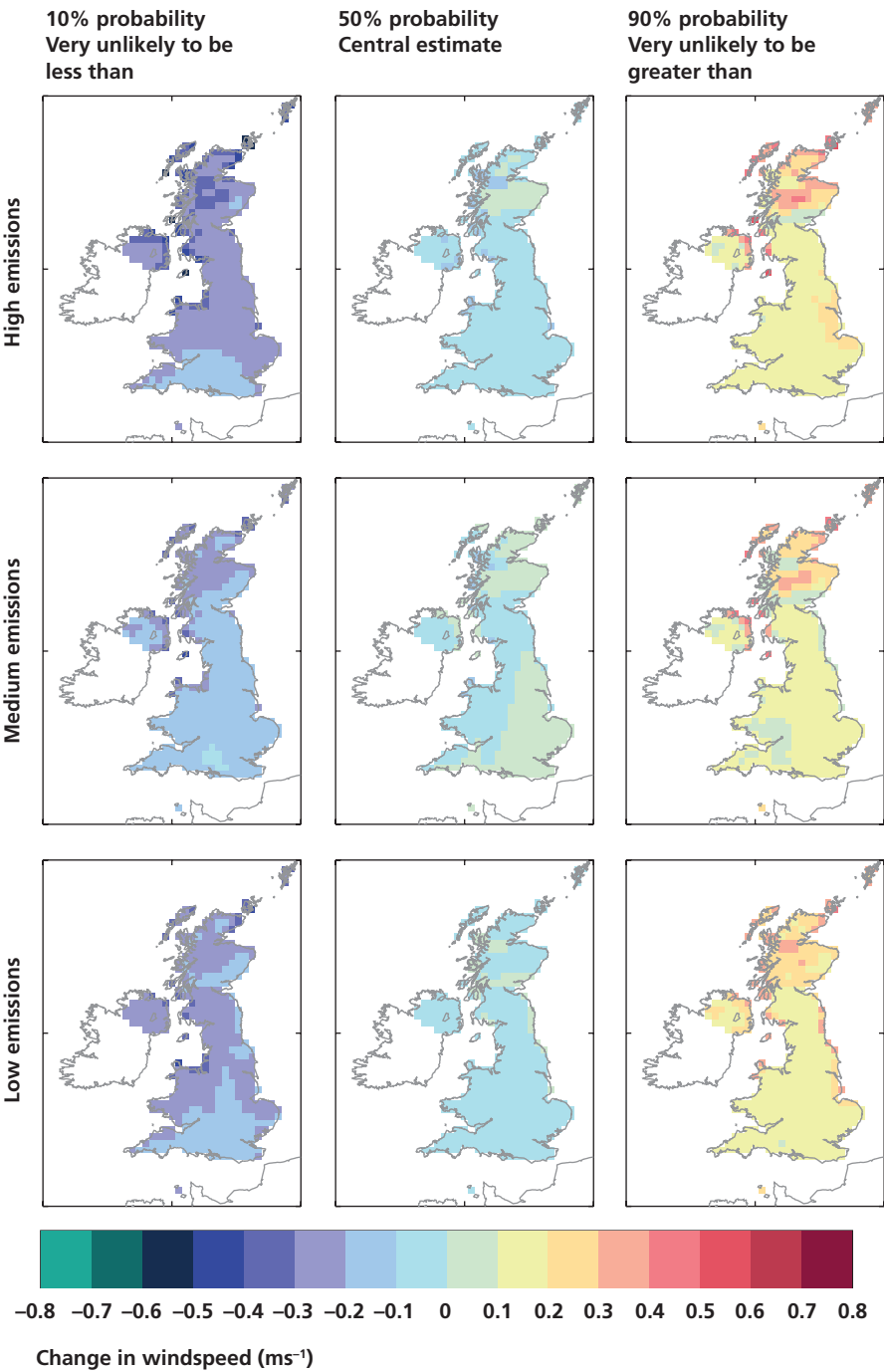


Figure 6: 10, 50, and 90% probability level maps of changes in winter wind speed (ms⁻¹) for the 2050s (2040–2069) relative to 1961–1990 for the Low (bottom row), Medium (middle row), and High (top row) emission scenarios.

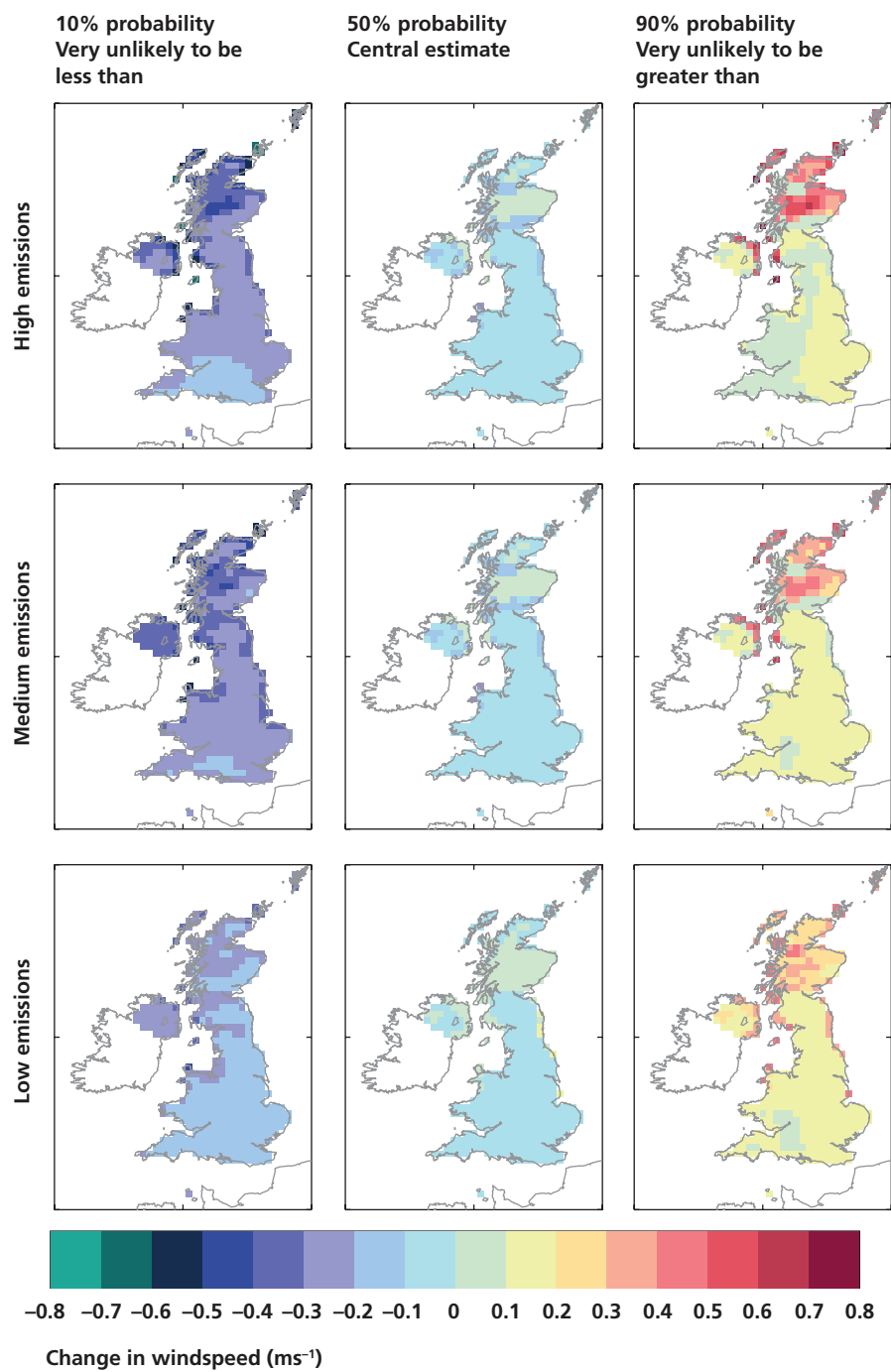


Figure 7: 10, 50, and 90% probability level maps of changes in winter wind speed (ms^{-1}) for the 2080s (2070–2099) relative to 1961–1990 for the Low (bottom row), Medium (middle row), and High (top row) emission scenarios.

Acknowledgements

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